AaharSetu

Bridging Food to Lives

SOLUTION APPROACH & DESIGN

Team Name: Quantum crew

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1. Problem Statement

India discards approximately 67 million tonnes of food every year — enough to feed 200 million people. Meanwhile, 189 million Indians go to bed hungry. The gap is not a food shortage but a failure of last-mile distribution.

67M	189M	40%	₹92K
Tonnes wasted annually	Indians facing hunger daily	Food wasted before use	Crore annual economic loss

2. Solution Overview

AaharSetu is a multi-role web platform that automates the entire food-rescue pipeline — connecting restaurants, NGOs, and volunteers in real time.

- Connects restaurants, NGOs, and volunteers on a single platform
- Real-time food redistribution with live status tracking
- Location-based smart matching for faster pickups
- End-to-end workflow tracking with feedback loop

3. User Personas

RESTAURANT	**Ravi — Restaurant Manager, Chennai** **Pain Point:** Has 15–20 kg of leftover food every evening with no structured way to donate. **Goal:** List food quickly, reach the right NGO, and get pickup confirmation.
NGO	**Sunita — NGO Coordinator, Bengaluru** **Pain Point:** Struggles to coordinate pickups across multiple donors without a system. **Goal:** View nearby food, accept in one tap, and track volunteer assignments.
VOLUNTEER	**Arjun — College Student, Mumbai** **Pain Point:** Wants to contribute but doesn't know when food is available or where to go. **Goal:** Get clear assignments, navigate easily, and log completion.
ADMIN	**Priya — Platform Administrator** **Pain Point:** Needs system-wide visibility to ensure quality and resolve issues. **Goal:** Monitor all donations, volunteers, and NGOs from one dashboard.

4. System Workflow

The platform follows a clear five-step end-to-end flow:

1. Restaurant uploads surplus food with photo, quantity, and expiry time
2. Nearby NGOs are notified; an NGO accepts the listing
3. A volunteer is automatically assigned based on proximity
4. Volunteer picks up food and marks it in-transit; NGO sees live ETA
5. NGO confirms delivery; both parties submit feedback

Restaurant → **NGO** → **Volunteer** → **Delivery** → **Feedback**

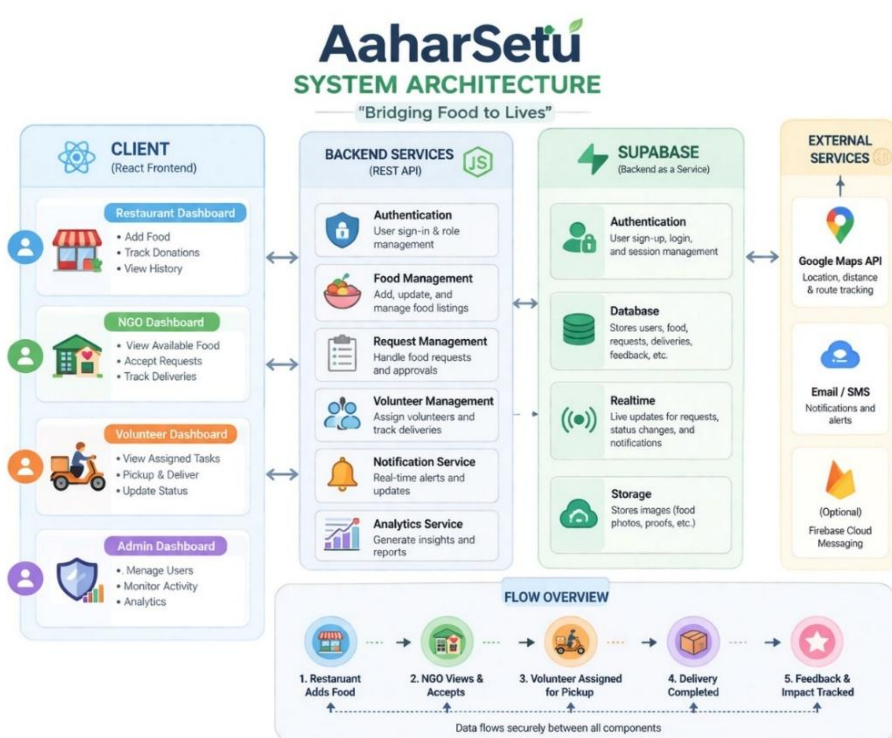
AaharSetu System Flow

5. Architecture Overview

- Frontend: React.js — responsive role-based dashboards
- Backend: Node.js / Express.js — RESTful API with JWT authentication
- Database: Supabase (PostgreSQL) — real-time subscriptions + Row-Level Security
- Maps: Google Maps API — geolocation, routing, and distance matching

Frontend React.js	→	**Backend** Node.js APIs	→	**Database** Supabase

System Architecture Diagram



6. Modules & Features

Restaurant Module

- Add food listing with photo, quantity, type, and expiry
- Track donation status and receive pickup confirmations

NGO Module

- Map view of nearby available food listings
- One-tap acceptance with volunteer auto-assignment
- Real-time volunteer tracking and ETA display

Volunteer Module

- View assigned tasks with restaurant location and navigation
- Update status: Accepted → Picked Up → Delivered

Admin Module

- Global dashboard: total donations, active volunteers, NGOs served
- User management, dispute resolution, and analytics reports

7. Tech Stack

Category	Technology
Frontend	React.js + Tailwind CSS
Backend	Node.js + Express.js
Authentication	Supabase Auth (JWT)
Database	Supabase (PostgreSQL + Realtime)
Maps	Google Maps API
Hosting	Vercel (frontend) + Render (backend)

8. MVP Features & Sprint Plan

- Role-based login and registration
- Food donation listing with photo upload
- NGO map-based browser and accept action
- Volunteer assignment and GPS navigation
- Live status tracking and post-delivery feedback
- Admin monitoring dashboard